Questions

Q1.

(a) Find the first four terms, in ascending powers of x, of the binomial expansion of

$$\left(2-\frac{1}{4}x\right)^6$$

(4)

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(b) Given that x is small, so terms in x^4 and higher powers of x may be ignored, show

$$\left(2-\frac{1}{4}x\right)^6+\left(2+\frac{1}{4}x\right)^6=a+bx^2$$

where a and b are constants to be found.

(3)

(Total for question = 7 marks)

(Q04 WMA12/01, June 2019)

Q2.

(a) Find the first 4 terms, in ascending powers of x, in the binomial expansion of

$$\left(1+\frac{x}{4}\right)^{12}$$

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giving each coefficient in its simplest form.

(3)

(b) Find the term independent of x in the expansion of

$$\left(\frac{x^2+8}{x^5}\right)\left(1+\frac{x}{4}\right)^{12}$$

(3)

(Total for question = 6 marks)

(Q03 WMA12/01, Oct 2019)

Q3.

One of the terms in the binomial expansion of $(3 + \alpha x)^6$, where α is a constant, is $540x^4$

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(a) Find the possible values of α.

(4)

(b) Hence find the term independent of x in the expansion of

$$\left(\frac{1}{81} + \frac{1}{x^6}\right)(3 + \alpha x)^6$$

(3)

(Total for question = 7 marks)

(Q02 WMA12/01, Jan 2020)

Q4.

A curve C has equation $y = f(x)$ where

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$$f(x) = (2 - kx)^5$$

and k is a constant.

Given that when $f(x)$ is divided by $(4x - 5)$ the remainder is $\frac{243}{32}$

(a) show that $k = \frac{2}{5}$

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(2)

(b) Find the first three terms, in ascending powers of x, of the binomial expansion of

$$\left(2-\frac{2}{5}x\right)^5$$

giving each term in simplest form.

(3)

Using the solution to part (b) and making your method clear,

(c) find the gradient of C at the point where $x = 0$

(2)

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(Total for question = 7 marks)

(Q02 WMA12/01, Oct 2022)

Q5.

(a) Find the first 4 terms, in ascending powers of x, of the binomial expansion of

$$\left(2-\frac{x}{4}\right)^{10}$$

giving each term in its simplest form.

(4)

(b) Hence find the constant term in the series expansion of

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$$\left(3-\frac{1}{x}\right)^2\left(2-\frac{x}{4}\right)^{10}$$

(3)

(Total for question = 7 marks)

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(Q01 WMA12/01, Oct 2020)

Q6.

(a) Find the first three terms, in ascending powers of x, of the binomial expansion of

$$(2 + px)^6$$

where p is a constant. Give each term in simplest form.

(4)

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Given that in the expansion of

$$\left(3 - \frac{1}{2}x\right)(2 + px)^6$$

the coefficient of x^2 is $-\frac{3}{4}$

(b) find the possible values of p.

(4)

(Total for question = 8 marks)

(Q04 WMA12/01, Jan 2021)

Q7.

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(a) Find, in ascending powers of x, up to and including the term in x^3, the binomial expansion of

$$\left(2 + \frac{x}{8}\right)^{13}$$

fully simplifying each coefficient.

(4)

(b) Use the answer to part (a) to find an approximation for 2.0125^{13}

Give your answer to 3 decimal places.

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(3)

Without calculating 2.0125^{13}

(c) state, with a reason, whether the answer to part (b) is an overestimate or an underestimate.

(1)

(Total for question = 8 marks)

(Q04 WMA12/01, June 2021)

Q8.

(i) (a) Find, in ascending powers of x, the 2nd, 3rd and 5th terms of the binomial expansion of

$$(3 + 2x)^6$$

(3)

For a particular value of x, these three terms form consecutive terms in a geometric series.

(b) Find this value of x.

(3)

(ii) In a **different** geometric series,

- the first term is $\sin^2\theta$
- the common ratio is $2\cos\theta$
- the sum to infinity is $\frac{8}{5}$

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(a) Show that

$$5\cos^2\theta - 16\cos\theta + 3 = 0$$

(3)

(b) Hence find the exact value of the 2nd term in the series.

(3)

(Total for question = 12 marks)

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(**Q10 WMA12/01, Oct 2023**)

Q9.

$$f(x) = \left(2 + \frac{kx}{8}\right)^7$$ where k is a non-zero constant

(a) Find the first 4 terms, in ascending powers of x, of the binomial expansion of f(x).

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Give each term in simplest form.

(4)

Given that, in the binomial expansion of f(x), the coefficients of x, x^2 and x^3 are the first 3 terms of an arithmetic progression,

(b) find, using algebra, the possible values of k.

(Solutions relying entirely on calculator technology are not acceptable.)

(3)

(Total for question = 7 marks)

(**Q03 WMA12/01, Jan 2023**)

Q10.

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The first three terms, in ascending powers of x, of the binomial expansion of $(1 + kx)^{16}$ are

$$1, \quad -4x \quad \text{and} \quad px^2$$

where k and p are constants.

(a) Find, in simplest form,

(i) the value of k

(ii) the value of p

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(3)

$$g(x) = \left(2 + \frac{16}{x}\right)(1 + kx)^{16}$$

Using the value of k found in part (a),

(b) find the term in x^2 in the expansion of g(x).

(3)

(Total for question = 6 marks)

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(**Q01 WMA12/01, Oct 2021**)

Q11.

(a) Find the first 4 terms, in ascending powers of x, of the binomial expansion of

$$\left(2 - \frac{kx}{4}\right)^8$$

where k is a non-zero constant. Give each term in simplest form.

(4)

$$f(x) = (5 - 3x)\left(2 - \frac{kx}{4}\right)^8$$

In the expansion of f(x), the constant term is 3 times the coefficient of x.

(b) Find the value of k.

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(3)

(Total for question = 7 marks)

(**Q03 WMA12/01, Jan 2022**)

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Q12.

Find the first four terms, in ascending powers of x, of the binomial expansion of

$$\left(2+\frac{3}{8}x\right)^{10}$$

Give each coefficient as an integer.

(Total for question = 4 marks)

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(Q01 WMA12/01, June 2022)

Q13.

The binomial expansion, in ascending powers of x, of

$$(3 + px)^5$$

where p is a constant, can be written in the form

$$A + Bx + Cx^2 + Dx^3 \ldots$$

where A, B, C and D are constants.

(a) Find the value of A

(1)

Given that

- $B = 18D$
- $p < 0$

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(b) find

(i) the value of p

(ii) the value of C

(6)

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(Total for question = 7 marks)

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